

c) The standard matrix of a linear transformation from $\mathbb{R}^2$ to $\mathbb{R}^2$ that reflects points through the horizontal axis, the vertical axis, or the origin has the form $\begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}$ where a and d are ± 1 .

Solution: From the Reflections Table on Page 74, we know that the given statement is True.

d) A mapping $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is one-to-one if each vector in $\mathbb{R}^n$ maps onto a unique vector in $\mathbb{R}^m$.

Solution: By the defn. of one-one mapping on Page 76, we know that the given statement is False.

e) If A is a 3×2 matrix, then the transformation $x \mapsto Ax$ cannot map $\mathbb{R}^2$ onto $\mathbb{R}^3$.

Solution: By definition, the pivot position of a matrix corresponds to the leading 1 of a row in rref. By Theorem 1 and Row Reduction algorithm, the matrix A can be converted to a unique rref. From the defn. of rref, we know that there can exist at most 2 rows of A which has pivots. Since every row does not have a pivot, the columns of A do not span $\mathbb{R}^3$ (By Theorem 4). By Theorem 12(a), the transformation cannot map $\mathbb{R}^2$ onto $\mathbb{R}^3$.

$\therefore$ The given statement is True $\square$

In Exercises 25-28, determine if the specified linear transformation is (a) one-to-one and (b) onto. Justify each answer.

~~By Theorem 12(b), a transformation T is linear if and only if the columns of its standard matrix A are linearly independent.~~
By defn. of Ax , the columns of A are linearly independent if and only if the matrix equation $Ax=0$ has only the trivial solution.
By Theorem 3, the matrix equation $Ax=0$ has the same solution set as the system of equations corresponding to the augmented matrix $[A \ 0]$.